MODULE 06

Environmental Protection & Ethics

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Impact on Construction Industry on Global Warming

Carbon Emissions

- **Cement production:** Making cement releases a lot of carbon dioxide (CO_2). In fact, cement is responsible for about **7–8% of all global CO_2 emissions**. This is because when limestone is heated to make cement, it naturally gives off CO_2.
- **Steel production:** Making steel also requires a lot of energy, usually from burning coal, which adds more emissions.
- **Construction machinery:** Excavators, cranes, trucks, and generators at sites run on diesel, producing CO_2, nitrous oxides (NO_x), and fine dust particles.
- **Transport:** Construction materials are often carried over long distances by trucks or ships, which also burn fuel and add to global warming.

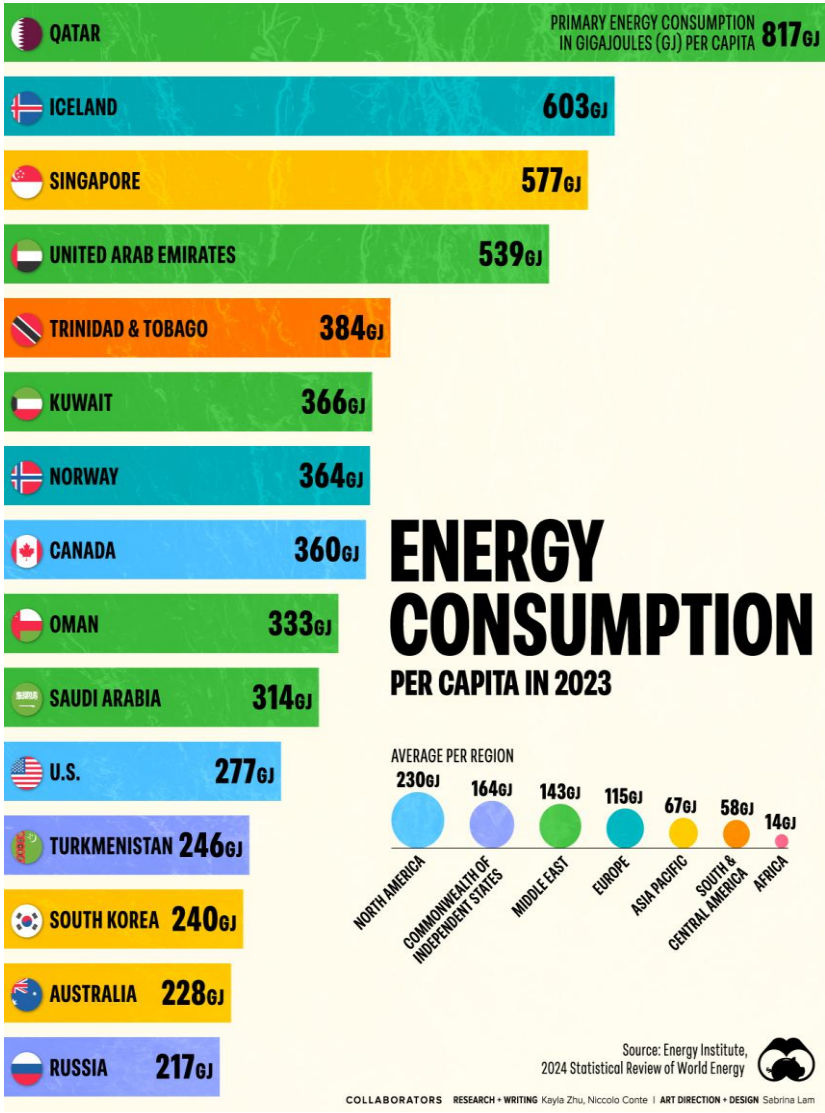
Impact: All these activities increase the level of greenhouse gases in the air, trapping heat and making the Earth warmer.

Impact on Construction Industry on Global Warming

High Energy Consumption

- Buildings use around **30–40% of the world's total energy**.
- Energy is required during construction (for machinery and material production) and later during the building's life (for lighting, cooling, heating, lifts, etc.).
- In many countries, electricity still comes from coal, oil, or gas. So, every time we use electricity in a building, more CO_2 is released into the atmosphere.

Impact: This leads to higher global warming and puts more pressure on energy resources.



Impact on Construction Industry on Global Warming

Overuse of Natural Resources



- Construction depends heavily on natural resources like **sand, gravel, stones, clay, wood, and water**.
- **Sand mining** destroys riverbeds and coastal areas.
- **Deforestation** occurs when trees are cut to make space for new projects or to supply timber. This reduces the number of trees that can absorb carbon.



- **Water use:** Making concrete and curing it needs a lot of fresh water, which is already limited in many regions.

Impact: *The Earth loses natural "carbon sinks" (like forests) and faces scarcity of water and minerals.*

Impact on Construction Industry on Global Warming

Waste Generation

- Construction and demolition (C&D) waste makes up about **25–30% of all waste produced worldwide**.
- Materials like broken concrete, bricks, plaster, wood, and metal are often dumped in landfills instead of being recycled.
- Some wastes, like paints and solvents, are hazardous and can poison soil and water.
- In landfills, waste can release **methane gas**, which is 25 times more harmful than CO_2 in trapping heat.

Impact: *Poor waste management increases pollution and worsens climate change.*

Impact on Construction Industry on Global Warming

Water Pollution

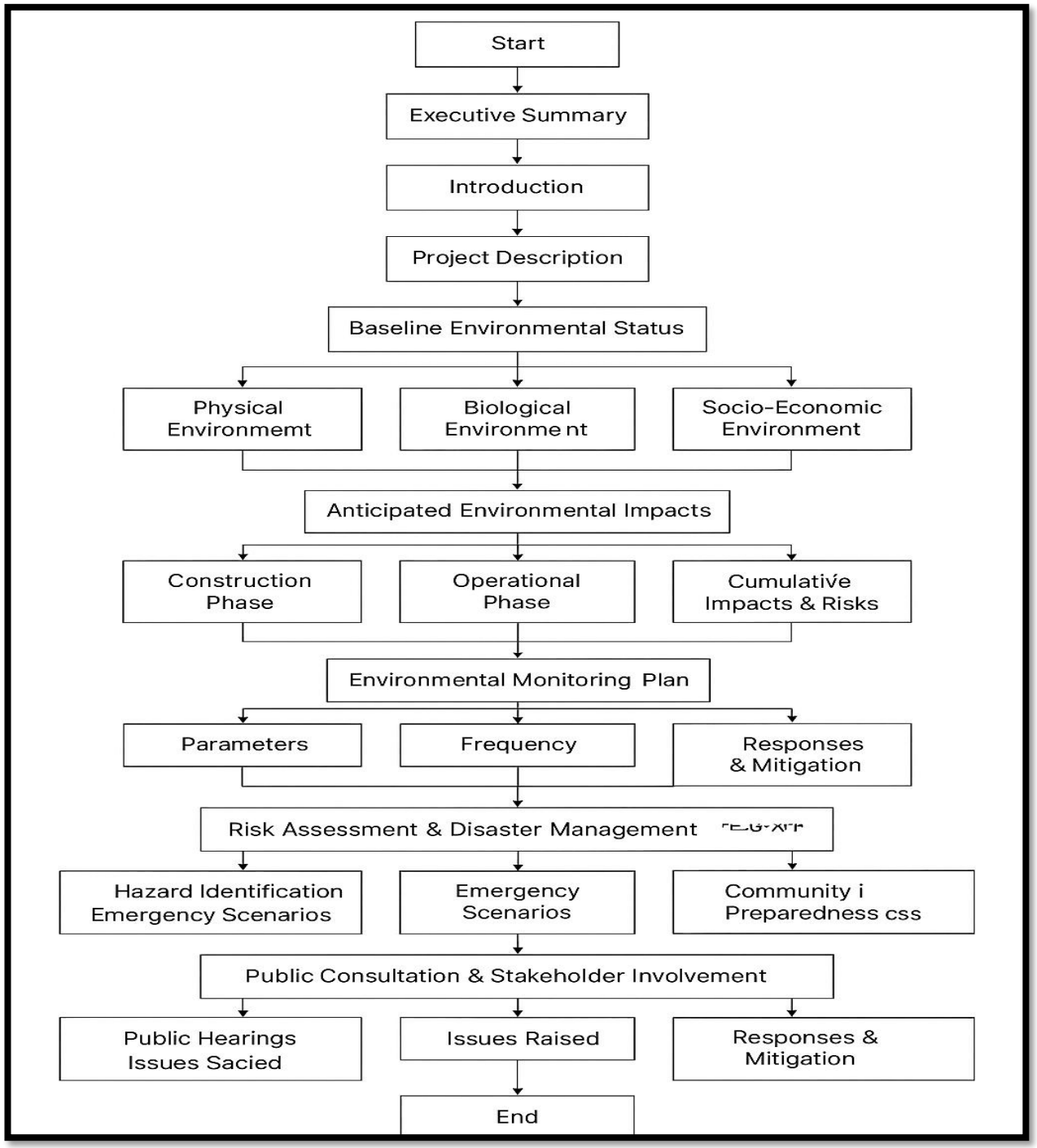
- Construction sites often wash away cement slurry & chemicals into rivers and lakes.
- Large coastal projects damage marine ecosystems by dredging and dumping waste.
- Polluted water affects fish, plants, and coral reefs, reducing the ability of oceans to absorb CO_2.

Impact: *Oceans, which are the biggest natural absorbers of carbon, become weaker in fighting global warming.*

Indirect Impacts

- **Poor design of buildings:** If buildings are not energy-efficient, they waste electricity for many years.
- **Construction in risky areas:** Building in flood zones or coastal areas increases vulnerability to climate change disasters.
- **Long life cycle impact:** A building continues to affect the environment throughout its life — in its use of electricity, water, and maintenance.

EIA REPORT FLOWCHART





Navi Mumbai International Airport (NMIA) EIA Report Overview

Overview of NMIA Project

- Greenfield international airport at Ulwe, Navi Mumbai (MMR)
- Total area: ~1,160 hectares
- Constructed in phases – final capacity ~90 million passengers/year
- Large cargo handling capacity
- Includes runways, terminals, cargo, road/transport connectivity

Key Environmental & Social Issues

Mangroves and Wetlands Loss

- The airport site involves low-lying coastal area with mangroves, mudflats, water bodies etc.
- Estimated loss of about 161.50 hectares of mangrove.
- Mangrove loss is particularly sensitive because they serve multiple ecological functions: biodiversity (birds etc.), buffering against storm surges, preventing coastal erosion, maintaining hydrology.

Key Environmental & Social Issues

Bird-Strike Risk / Avifauna

- The project area hosts a large number of bird species, including those that rely on mangrove and wetland habitats.
- Proximity to the Karnala Bird Sanctuary is a concern.
- The EIA report initially claimed the sanctuary is 12.5 km away, but activists state the actual distance is approximately 9 km.
- Projects within 10 km of a protected area typically require special clearance from the National Board for Wildlife (NBWL).
- Bird-related risks include:
  - ❖ **Safety risk:** Potential bird strikes affecting aircraft operations.
  - ❖ **Conservation risk:** Impact on migratory and resident bird species.

Key Environmental & Social Issues

River Diversion / Hydrology

- The project involves diversion/training of Gadhi River and Ulwe River.
- Concerns exist regarding impacts on flood risk, water flow, and drainage.
- Project reports claim that the diversions are “essential, cost-effective, and less environmentally damaging,” but this is debated.

Land Acquisition, Displacement, Rehabilitation

- Multiple villages are affected, resulting in loss of land, homes, and livelihoods.
- Project Affected Persons (PAPs) have raised complaints about:
  - Insufficient compensation.
  - Problems with records of land ownership.
  - Issues with the quality and location of resettlement land.

Key Environmental & Social Issues

Regulatory / Legal / Clearances

- Clearance under Coastal Regulation Zone (CRZ) rules had to be amended to allow conversion of mangrove forest for the airport.
- Wildlife clearance is required due to proximity to a wildlife sanctuary; controversies exist regarding whether NBWL clearance was obtained and whether proper measurements were used.
- Issues were noted with EIA transparency and stakeholder participation, including protests and boycotted public hearings.

Environmental Mitigation / Compensation Measures

- Compensatory mangrove plantation is proposed; the compensatory forestry "shall be double the area of loss of mangroves."
- Certain designs of river diversions are considered to be relatively less environmentally damaging.

Key Findings / Conclusions

- The NMIA site is considered the most feasible, balancing infrastructure needs and minimizing environmental damage.
- River diversions of Gadhi and Ulwe are hydrologically manageable and not expected to cause major disruptions.
- Ecological impacts are significant but can be managed with mitigation and compensation measures, such as compensatory mangrove plantations and careful river diversion design.
- Overall, the project is feasible and relatively sustainable if proper environmental management is applied.

Gaps / Criticisms / Potential Risks

- **Accuracy / Distance Disputes:** The claimed distance to Karnala Bird Sanctuary is disputed; if less than 10 km, stricter regulatory scrutiny is required.
- **Adequacy of Compensation:** Compensatory mangrove plantations may not fully replicate the ecological functions of lost mangroves, especially if planted at distant sites, leading to local loss of ecosystem services.
- **Public Participation / Transparency:** Public hearings were not always effective; some affected people were dissatisfied, and record-keeping was problematic.
- **Long-term Monitoring:** Uncertainty exists about the robustness of post-project monitoring for biodiversity, hydrology, and bird strike risk.
- **Cumulative Impacts:** Urban expansion, road traffic, and other developments may increase the overall environmental burden beyond initial projections.

Mitigation & Management Measures (Recommended)

- **Compensatory Mangrove Reforestation:** Plant mangroves in or near impacted zones, possibly double the area lost.
- **River Diversion / Training:** Implement hydraulic designs that preserve drainage and reduce flood risk.
- **Bird Hazard Management:** Conduct studies to map bird habitats and flight paths, avoid bird attractants, and modify designs to reduce bird strike risk.
- **Rehabilitation & Resettlement:** Provide affected villagers with land titles, fair compensation, and suitable relocation sites.
- **Legal Clearances:** Obtain all necessary approvals from environmental, wildlife, and coastal regulation authorities.
- **Monitoring & Reporting:** Conduct regular environmental monitoring of air, noise, hydrology, and biodiversity, with periodic audits.

Paris Agreement 2020 – Key Highlights

Introduction

- Landmark global treaty to combat climate change
- Adopted in **2015;** aims to limit global warming well below 2°C, striving for 1°C.
- 195 countries are parties, including India

Key Objectives

- Limit global temperature rise to <2°C, idéally **1.5**°C
- Climate finance: Developed countries support developing nations for mitigation & adaptation
- **Climate Finance Milestone:** 2022 – USD 115.9 billion mobillized, surpassing USD 100B goal

2020–2025 Developments

- **U.S. Withdrawal & Rejoining:** Left in 2020 (Trump), rejoined in 2021 (Biden), recent withdrawal order in 2025
- **China's Targets:** Reduce GHG emissions 7–10% by 2035; EU criticized as insufficient
- **Climate Finance Milestone:** 2022 – USD 115.9 billion mobilized, surpassing USD 100B goal

Conclusion

- Paris Agreement remains a pivotal framework for climate action
- Success depends on global commitment & cooperation

India's Climate Target as per Paris Agreement 2020

Category	Target (vs. 2005 levels)	Key Strategy
GHG Emission Intensity	Reduce intensity of GHG by 33-35%	Focus on low-carbon economic growth over absolute cuts.
Renewable Energy (RE)	Achieve 40% of cumulative electricity capacity from non-fossil fuels.	Target 450 GW of total RE capacity (Solar, Wind, etc.).
Carbon Sink	Absorb 2.5 - 3 billion tonnes of equivalent.	Enhance afforestation and sustainable forest management.

Supporting Action Plans

- **Adaptation & Resilience:** Strengthen **climate-resilient agriculture**, promote water efficiency, sustainable urban planning, and disaster preparedness.
- **National Policies:** Implement the **National Action Plan on Climate Change (NAPCC)**, fostering energy efficiency, clean transportation, and green industries.

Ethical Responsibilities of Stakeholders for Environmental Protection

Civil Engineers

- **Design for minimal harm:** Create plans that reduce long-term negative impacts on the environment.
- **Prioritize sustainability:** Balance current construction needs with the ability of the planet to support future generations.
- **Choose eco-friendly materials:** Use recycled, locally sourced, or low-impact resources wherever possible.
- **Assess environmental impacts:** Study and report how the project could affect natural systems, such as water bodies, wildlife habitats, and vegetation.
- **Advocate green solutions:** Ethically promote sustainable ideas and systems to clients, even if initial costs are higher.



Contractors

- **Comply with regulations:** Follow all federal, state, and local environmental laws and permit requirements.
- **Prevent pollution:** Implement measures to stop soil, debris, and chemicals from contaminating rivers, lakes, or storm drains.
- **Manage waste responsibly:** Develop plans to sort, recycle, and reduce construction debris sent to landfills.
- **Handle hazards safely:** Properly store, handle, and dispose of all hazardous materials, including fuels and chemicals.
- **Protect the site:** Physically safeguard sensitive areas, such as wetlands, mature trees, and other ecological zones.
- **Educate the workforce:** Train all site personnel in environmental protocols and best practices.



Other Parties (Owners, Suppliers, and Regulators)

- **Owners must invest adequately:** Allocate sufficient funds and time to ensure environmental and sustainable practices are implemented.
- **Owners must set clear goals:** Define environmental standards and sustainability certifications in contracts.
- **Suppliers must provide transparency:** Offer accurate information about the environmental impact of materials and products.
- **Subcontractors must comply:** Follow the environmental protection plans set by the general contractor.
- **Regulators must enforce rules:** Consistently and fairly enforce environmental laws to protect public trust and the ecosystem.



THANK YOU